Noman Saffat Sajid

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Education

North South University September 2021 - September 2025

Bachelor of Science in Computer Science and Engineering (BSCSE) ***CGPA:*** *3.87 out of 4.00*

- **Relevant Coursework:** Machine Learning, Pattern Recognition, Design and Analysis of Algorithms, Data Structures and Algorithms, Linear Algebra, Probability and Statistics, Numerical Methods, Calculus and Analytical Geometry

Research Experience

Developed a preprocessing module to reduce deep learning model cost through cropping 2024 - Ongoing

Under the supervision of Dr. Md. Shahriar Karim, Assoc. Prof. ECE, NSU

- Developed a preprocessing module that reduces training time and computational expense (CO_2 emission, and energy consumption) by 30-50% across different deep learning architectures, with only a 1-3% decrease in accuracy.
- Developed a Principal Component Analysis based statistical algorithm for image cropping dimension selection.
- Contributed substantially to paper writing, figure creation, and experimental analysis of the proposed method.

Mathematical model for open set recognition September 2025 - May 2026

Under the supervision of Dr. Mohammad Shifat-E-Rabbi, Asst. Prof. ECE, NSU

- Performed benchmarking of proposed method against existing deep learning and machine learning methods, and demonstrated competitive or superior performance of the proposed method.
- Led the paper writing, figure creation, and experimental analysis of the proposed method.

Publications

Peer-Reviewed Publications

N. S. Sajid, M. R. Khan, and M. S. Karim. "Objects Cannot Hide Behind The Shades – We Can See Through and Localize." *European Conference on Computer Vision (ECCV) CDEL Workshop, 2026.* ***[Oral Presentation]***

N. S. Sajid and M. S. Karim. "Target Localization in Cell-Based Image Analysis and Disease Diagnosis." *International Conference on Learning Representations (ICLR) LMRL Workshop, 2025.*

F. M. A. Hossain, **N. S. Sajid**, and R. M. Rahman. "Semantic-Preserving Image Compression and Restoration Pipeline for Bandwidth-Constrained UAV Applications: A Task-Aware Evaluation Framework." *Artificial Intelligence and Applications (AIA), 2026.*

Under Review / In Preparation

N. S. Sajid, M. R. Khan, and M. S. Karim. "Objects Cannot Hide Behind The Shades – We Can See Through and Localize." *Manuscript in preparation for Transactions on Machine Learning Research (TMLR). (Extended version of ECCV 2026 Workshop paper).*

N. S. Sajid*, Nujhat Raisa Arpa*, Amrijit Biswas*, M. Rahman, M. Subha, M. Shifat-E-Rabbi, S. Rahman, and G. K. Rohde. "A Sliced-Wasserstein Distance Based Approach for Open Set Recognition." *Manuscript in preparation for Pattern Recognition Letters (PRL). (*Shared first author)*

M. Rahman, M. Subha, **N. S. Sajid**, O.-I. Shahid, and M. Shifat-E-Rabbi. "Geometric Deformation Invariance in Data-Efficient Point Cloud Classification Using Linear Optimal Transport Subspaces." *Submitted to Pattern Recognition Letters (PRL).*

Course Projects

Image Processing Pipeline for Efficient Transmission of UAV Images | *Senior year project*

- Designed and implemented a pipeline that uses color quantization as a compression strategy for faster image transmission, which achieves 68% file size reduction and 3× faster transmission times.
- Implemented a deep learning model that restores the image after transmission, from quality degradation caused by color quantization.

IoT-Based Smart Home Security and Monitoring System | *Embedded Systems Course Project*

- Interfaced the STM32 blue pill microcontroller with bluetooth module, OLED display and keyboard.
- Implemented a mobile app using MIT app inventor, initialized a database, and interfaced with the Bluetooth module.

Work Experience

Research Assistant (RA) September 2025 - May 2026

Under Dr. Mohammad Shifat-E-Rabbi, Asst. Prof. ECE, NSU

- Worked on a mathematical model for open set recognition

Extracurricular Experience

DB Hackathon: Database Design Challenge October 2023

2nd prize winner *NSU, Dhaka*

- Solved different situation-based problems using Entity-Relationship diagrams.
- Wrote complex SQL queries using joins, subqueries, and aggregate functions, to solve competition problems.

Skills

Languages: English and Bengali
Programming Languages: Python (OpenCV, PyTorch, NumPy, Matplotlib), C and Java
Operating Systems: Linux and Windows